A geometric characterization of the standard MOND interpolating function: $\mu(x) = x/\sqrt{1+x^2}$ as the isotropic distribution of $\cot\theta$

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Abstract. The interpolating function $\mu(x)$ of Modified Newtonian Dynamics (MOND) phenomenology has been chosen empirically for four decades; standard reviews state explicitly that no derivation from first principles exists. We point out an exact identity: the standard interpolating function, $\mu(x) = x/\sqrt{1+x^2}$, is the cumulative distribution of $\cot\theta$ for directions distributed isotropically on the hemisphere — equivalently, the threshold density it implies, $\mu'(\Lambda_c) = (1+\Lambda_c^2)^{-3/2}$, is exactly the isotropic probability density of the ratio of longitudinal to transverse field projections. Operationally, $\mu(x)$ is the solid-angle fraction of channels activated by the tilt criterion $|\mathbf{g} \times \hat{n}| > a_0|\hat{n} \cdot \hat{g}|$: misalignment with the local field beyond $\arctan(a_0/g)$. The statement is elementary spherical geometry with no free parameters; a_0 enters only as the unit of the threshold. We delimit the content precisely: any monotone interpolating function is *some* solid-angle fraction (the correspondence gate $\leftrightarrow$ μ is one-to-one), so the specificity of the standard μ is that its threshold variable is the canonical projection ratio $\cot\theta$, and its gate is a linear comparison of field projections — the only such gate among the simple projection criteria (tilt, magnitude, longitudinal, perpendicular) with the correct MOND limits, and the one that rigidly fixes the member $\alpha = 2$ of the empirical family $\mu_\alpha = x/(1+x^\alpha)^{1/\alpha}$. As corollaries, the flow function $\beta(\Phi) = -\Phi(2+\Phi)(1+\Phi)/[2+\Phi(2+\Phi)]$ follows exactly, with attractor slope $\beta'(0) = -1$ and curvature $\kappa = 1/2$, making the baryonic Tully–Fisher relation a dynamical attractor. A Monte Carlo over 10^7 hard gates reproduces $\beta'(0) = -0.99$, $\kappa = 0.47$ with no tuning. All symbolic and numerical results are reproducible from the scripts archived with this letter.

Keywords: MOND; interpolating function; isotropy; solid angle; emergent gravity; radial acceleration relation; baryonic Tully–Fisher relation

1 Introduction

MOND phenomenology [Milgrom 1983a,b,c] accounts for galactic rotation curves through a relation $\mu(g/a_0)\,\mathbf{g} = \mathbf{g}_N$ (or its inverted form), where the interpolating function $\mu(x)$ satisfies $\mu \to x$ for $x \ll 1$ and $\mu \to 1$ for $x \gg 1$, and $a_0 \simeq 1.2 \times 10^{-10}\ \mathrm{m\,s^{-2}}$. The functional form of μ has always been an empirical input: Milgrom (1983a) proposed the simple $(x/(1+x))$ and standard $(x/\sqrt{1+x^2})$ forms as convenient examples; AQUAL [Bekenstein & Milgrom 1984] requires only the asymptotic limits; and the Living Review by Famaey & McGaugh (2012) states that no derivation of μ from first principles is available. The radial-acceleration relation [McGaugh, Lelli & Schombert 2016a] measured on the SPARC sample [Lelli, McGaugh & Schombert 2016b] has sharpened the empirical constraints on μ without changing its status as an ansatz.

This letter proves an elementary identity which, to our knowledge, has not appeared in the literature: **the standard interpolating function is the cumulative distribution of** $\cot\theta$ **under isotropy.** If gravitational "channels" are distributed isotropically in direction and a channel is active precisely when its misalignment with the local field exceeds a critical tilt $\arctan(a_0/g)$, then the active fraction is $\mu(x) = x/\sqrt{1+x^2}$ with $x = g/a_0$, exactly. No parameter is adjusted; the computation is a one-line integral over a spherical cap. We are equally explicit about what is *not* special: any monotone interpolating function can be written as a solid-angle fraction under some angular gate, because the correspondence between monotone gates and monotone μ's is one-to-one (§3). The non-trivial content is *which* threshold variable the standard function singles out — the ratio of longitudinal to transverse projections, $\cot\theta$, the canonical dimensionless coordinate of a direction relative to the field — and that the corresponding gate is a linear comparison of field projections, unlike the gates implied by the other members of the empirical μ_α family.

We emphasize the scope honestly. The theorem is purely geometric: it converts the choice of μ into the choice of an activation criterion, and shows that criterion to be uniquely selective within a natural class. The physical origin of the tilt criterion — why dissipation competes in precisely this projection — is a separate dynamical question that we take up in a forthcoming companion paper, building on the emergent-gravity framework already presented in [Figueroa Torres 2026]; it is not required for any statement made here. Likewise, the value of a_0 is not derived here; it enters only as the unit of the threshold.

2 Axioms and theorem

Let $\mathbf{g}$ be the local gravitational field, $g = |\mathbf{g}|$, $x \equiv g/a_0$, and let $\hat{n}$ denote the direction of a channel (the microscopic degree of freedom whose activation is at stake; its nature is not needed for the theorem).

Axiom 1 (isotropy)**.** *Channels carry unoriented directions (axes): $\hat{n}$ and $-\hat{n}$ label the same channel. The space of channel directions is therefore the hemisphere $\hat{n}\cdot\hat{g} \geq 0$ — not half of the possibilities but all of them — with the rotation-invariant (Haar) measure $\sin\theta\, d\theta\, d\phi$, where $\theta \in [0, \pi/2]$ is the angle between $\hat{n}$ and $\hat{g}$. (Equivalently, on the full sphere the criterion below is written with $|\hat{n}\cdot\hat{g}|$ and every statement is unchanged.)*

Axiom 2 (tilt activation)**.** *A channel is active if and only if*

$$|\mathbf{g} \times \hat{n}| > a_0 |\hat{n}\cdot\hat{g}|, \tag{1}$$

equivalently $g\sin\theta > a_0\cos\theta$, i.e., misalignment beyond the critical cone

$$\theta_0(x) = \arctan(1/x). \tag{2}$$

Theorem. *Under Axioms 1–2, the active fraction of channels is*

$$\mu(x) = \frac{\Omega_{active}}{\Omega_{total}} = \int_{\arctan(1/x)}^{\pi/2} \sin\theta\, d\theta = \cos\big(\arctan(1/x)\big) = \frac{x}{\sqrt{1+x^2}}. \tag{3}$$

Proof. The measure of Axiom 1 normalizes to $\int_0^{\pi/2}\sin\theta\, d\theta = 1$ on the hemisphere (ϕ integrates out). Axiom 2 activates the band $\theta > \theta_0$, whose measure is $\cos\theta_0$ — the complement of the spherical cap of half-angle θ_0. With $\theta_0 = \arctan(1/x)$, $\cos\theta_0 = x/\sqrt{1+x^2}$. $\square$

Both MOND limits are immediate, and the small-x expansion is

$$\mu(x) = x - \tfrac{1}{2}x^3 + \mathcal{O}(x^5), \qquad \mu(x \to \infty) \to 1. \tag{4}$$

Two remarks. (i) At $g = a_0$ the critical cone is exactly $45°$: the acceleration scale marks the tilt at which alignment and misalignment are balanced. (ii) **The central identity.** Equivalently, $\mu(x)$ is the fraction of solid angle with $\cot\theta < x$, and the threshold density it implies is

$$P(\Lambda_c) = \mu'(\Lambda_c) = (1 + \Lambda_c^2)^{-3/2}, \qquad \int_0^\infty P\, d\Lambda_c = 1, \tag{5}$$

which is exactly the probability density of $\cot\theta$ for isotropic directions. This — not the bare statement "a solid-angle fraction" — is where the specific content lives: the standard interpolating function is the one whose hard thresholds are distributed as the projection ratio $\cot\theta = (\hat{n} \cdot \hat{g}\text{-component})/(\text{transverse component})$ of an isotropic ensemble, the canonical dimensionless coordinate of a direction relative to the field. Other interpolating functions correspond, through the same construction, to threshold variables that are not natural functions of the direction (§3).

3 Uniqueness within the class of simple projection gates

The tilt criterion is not one arbitrary choice among many. Consider the four simple activation criteria that can be built from the projections of $\mathbf{g}$ on $\hat{n}$ (all thresholds in units of a_0; all fractions computed with the measure of Axiom 1):

Gate	Criterion	Active fraction	Failure
T (tilt)	$g\sin\theta > a_0\cos\theta$	$x/\sqrt{1+x^2}$	— (both limits, analytic, monotonic)
M (magnitude)	$g > a_0\cos\theta$	x for $x \le 1$; 1 for $x > 1$	kink at $x = 1$; $\beta \equiv 0$, no attractor (see §4)
L (longitudinal)	$g\cos\theta > a_0$	0 for $x \le 1$; $1 - 1/x$ for $x > 1$	no deep-MOND limit ($\mu \equiv 0$ at low x)
P (perpendicular)	$g\sin\theta > a_0$	0 for $x \le 1$; $\sqrt{1 - 1/x^2}$ for $x > 1$	no deep-MOND limit

$$\mu_M(x \le 1) = x, \qquad \mu_L(x > 1) = 1 - \frac{1}{x}, \qquad \mu_P(x > 1) = \sqrt{1 - \frac{1}{x^2}}. \tag{6}$$

Only the tilt gate produces a μ that is analytic, strictly increasing, and linear in x as $x \to 0$.

We now state precisely how far uniqueness extends, and — equally important — how far it does not. For any activation rule "active iff $x > f(\theta)$" with f monotonically decreasing in the tilt angle, $\mu(x) = \cos(f^{-1}(x))$ under the isotropic measure. This correspondence is one-to-one: every monotone interpolating function is a solid-angle fraction under its own unique gate. It therefore selects nothing by itself — the simple function $x/(1+x)$, for instance, arises from the gate $f_1(\theta) = \cos\theta/(1-\cos\theta)$. What the bijection *does* deliver is a rigidity lemma: demanding the standard μ forces

$$f(\theta) = \cot\theta \qquad \text{(no freedom)}, \tag{7}$$

which is precisely criterion (1) — so any microscopic mechanism that produces the standard interpolating function through monotone angular gating must implement the tilt comparison, a strong constraint on model building. The selection content proper remains the four-gate result above, sharpened by a structural observation. The general member of the empirical family $\mu_\alpha(x) = x/(1+x^\alpha)^{1/\alpha}$ corresponds, through the bijection, to the threshold variable

$$f_\alpha(\theta) = \frac{\cos\theta}{(1-\cos^\alpha\theta)^{1/\alpha}}, \qquad f_2 = \cot\theta, \tag{7$'$}$$

and only for $\alpha = 2$ does the threshold variable equal the projection ratio of the direction — the longitudinal over the transverse component, $\cot\theta$ — so that the gate takes the two-projection form of criterion (1). Conversely, the tilt gate leaves no family freedom: it produces exactly the $\alpha = 2$ member. We deliberately do not claim uniqueness over broader classes of "simple" gates (e.g., arbitrary linear combinations of projections and magnitudes) without specifying such a class precisely; the four criteria of the table are the natural single-projection comparisons, and (7′) identifies what is structurally special about the standard member within the empirical family. Honest boundaries: the mirror orientation (activation when aligned) admits its own solution $\tilde{f} \neq \cot\theta$ and is excluded on physical rather than geometric grounds; rules that are non-monotone or depend on variables other than the tilt angle are outside the claim.

The failure of gate M is instructive and subtle: $\mu_M = x$ satisfies both MOND limits in a piecewise sense, yet it is dynamically empty — because $\mu_M - x\mu_M' \equiv 0$, the flow function of §4 vanishes identically and the Tully–Fisher relation does not attract. Matching asymptotic limits is not sufficient; the curvature of μ carries the dynamics.

4 Corollaries: the flow $\beta(\Phi)$ and the BTFR attractor

Within the screened scalar–tensor framework of [Figueroa Torres 2026], departures from the baryonic Tully–Fisher relation (BTFR) are governed by a flow in log-expansion time with flow function

$$\beta(u_0) = -\frac{u_0(\mu - u_0\mu')}{\mu(\mu + u_0\mu')}, \tag{8}$$

evaluated on the interpolating function. Substituting the theorem's μ and changing variable to $\Phi = \sqrt{1+u_0^2} - 1$ gives, exactly,

$$\beta(u_0) = -\frac{u_0^2\sqrt{1+u_0^2}}{2+u_0^2}, \qquad \beta(\Phi) = -\frac{\Phi(2+\Phi)(1+\Phi)}{2+\Phi(2+\Phi)} = -\Phi - \tfrac{1}{2}\Phi^2 + \mathcal{O}(\Phi^3). \tag{9}$$

Hence $\beta'(0) = -1$ (linear attractor: the BTFR is dynamically stable against perturbations) and quadratic curvature $\kappa = 1/2$, both now derived from Axioms 1–2 rather than postulated.

Two conventions are pinned explicitly to avoid ambiguity. First, all accelerations in this letter — including the BTFR departure variable Φ — are normalized to a_0, the threshold unit of Axiom 2. The slope $\beta'(0)$ and curvature κ are dimensionless ratios and are therefore invariant under a rescaling of that unit; only the location of the BTFR normalization is not. The framework paper [Figueroa Torres 2026] is written instead in units of its screening scale $g_{\mathrm{crit}} = a_0/4\pi$ — a factor-of-4π difference in the acceleration unit — and a correction propagating this rescaling through that paper is in preparation there; it does *not* affect any result derived in this letter, which is self-contained in a_0-units throughout. Second, with the flow parameter $\lambda = \ln(H/H_0)$ of [Figueroa Torres 2026,

§10], the near-fixed-point solution of (9) is $\Phi(\lambda) = \Phi_0 e^{-\lambda} = \Phi_0 H_0/H$: the fixed point is a UV attractor, and the intrinsic BTFR scatter tightens toward high redshift,

$$\sigma_\Phi(z) = \sigma_\Phi(0)\,\frac{H_0}{H(z)} \approx 0.023 \text{ dex at } z = 2 \text{ for } \sigma_\Phi(0) = 0.07 \text{ dex}, \tag{10}$$

assuming a flat ΛCDM background with $\Omega_m = 0.3$ (so that $H(z)/H_0 = \sqrt{\Omega_m(1+z)^3 + 1 - \Omega_m}$). We flag explicitly a distinction that is easy to conflate: (10) concerns the width around the BTFR and runs opposite to the zero-point evolution $V(z) \propto H(z)^{1/4}$, which rises with redshift — both are simultaneous consequences of the same drift of the acceleration scale. Equation (10) holds for the pure flow (fixed baryonic configuration); the observable scatter of real populations acquires in addition the evolution of $g_{\rm bar}(z)$, which lies outside the flow and is treated separately in the framework erratum.

Numerical verification (no tuning). A Monte Carlo ensemble of 10^7 hard gates with isotropic directions (thresholds $\Lambda_c = \cot\theta$, $\cos\theta$ uniform on $[0, 1]$) yields the empirical μ, μ' and, through (8), the ensemble β: point-by-point agreement with the closed form (9) at the level of the Monte Carlo noise (max. deviation 4.9×10^{-3}), with $\beta'(0) = -0.9926$ and $\kappa = +0.4739$ against pre-declared acceptance bands -1.00 ± 0.02 and $+0.50 \pm 0.05$. One finite-N systematic is declared: the $u \to 0$ region is ill-conditioned (the numerator $\mu - u\mu' \sim u^3$ competes with Monte Carlo noise), which biases the fitted κ slightly low; the point-by-point comparison is therefore restricted to $u \in [0.1, 1.2]$ and the conditioning analysis is documented in the archived scripts. As a foil, an exponential threshold distribution — equally normalized, not isotropic — returns $(\beta'(0), \kappa) = (-2.37, -7.24)$. Since β is a functional of μ, this is a sensitivity check on the pipeline rather than independent evidence for isotropy: the specificity of the isotropic ensemble is carried by identity (5), not by the Monte Carlo.

5 Discussion

What is claimed. (i) The standard MOND interpolating function is exactly the cumulative isotropic distribution of the projection ratio $\cot\theta$ — equivalently, the solid-angle fraction under the tilt gate (Theorem, eqs. (3) and (5)); (ii) since the correspondence gate $\leftrightarrow$ μ is one-to-one, this entails a rigidity lemma — any mechanism producing the standard μ through monotone angular gating must implement the tilt comparison (eq. (7)) — while the selection content proper is the four-gate result plus the structural fact that only the $\alpha = 2$ threshold variable is the projection ratio $\cot\theta$ (eqs. (6), (7′)); (iii) the BTFR attractor structure ($\beta'(0) = -1$, $\kappa = 1/2$) follows with no further input (eq. (9)).

What is not claimed. The theorem does not by itself derive MOND from first principles: it relocates the empirical content of μ into an activation rule and shows that rule to be geometrically distinguished. The dynamical origin of the tilt criterion (as a competition between a decoherence rate and a re-coherence rate, which also fixes the scale of a_0) is the subject of a forthcoming companion paper and is logically independent of the present result. The value of a_0 is an input here.

Relation to previous work. Derivations of an interpolating function from a physical mechanism do exist, and we state them precisely. Milgrom (1999) observed that the Unruh temperature excess of an accelerated observer in de Sitter space suggests $\hat\mu(x) = \sqrt{1 + (2x)^{-2}} - (2x)^{-1}$; Klinkhamer & Kopp (2011) and Klinkhamer (2012) obtained exactly that $\hat\mu$ — together with the Bekenstein–Milgrom modified Poisson equation and $a_0 = 4\pi(c/\hbar)k_B T_{\min}$ — from entropic gravity with a minimum temperature; Trippe (2013) obtained the simple function $x/(1+x)$ from a toy model of virtual massive gravitons; Pazy (2013) related μ to a ratio of holographic degrees of freedom without a closed form. Other frameworks obtain MOND-like behaviour without an interpolating function: emergent gravity yields $g_D = \sqrt{g_B a_0/6}$ [Verlinde 2017], superfluid dark matter postulates the MONDian phonon action $P(X) \propto X\sqrt{|X|}$ [Berezhiani & Khoury 2015], and quantum-gravity arguments recover the scale a_0 and the deep-MOND regime with the transition left unspecified [Smolin 2017]. None of these derives the standard $\mu(x) = x/\sqrt{1+x^2}$, and none proceeds by counting directions: the mechanisms are thermodynamic or field-theoretic, not solid-angle geometry. The present result is therefore new on two specific counts: it is the standard function that emerges, and it emerges from isotropy alone. That the entropic route selects $\hat{\mu}$ while direction counting selects the standard μ makes the transition region of the radial-acceleration relation a potential observational discriminant between the two mechanisms.

Falsifiability. The construction is rigid: isotropy plus a hard tilt threshold give exactly $x/\sqrt{1+x^2}$, not a family. Data preferring a materially different μ (e.g., the simple function at high accelerations) would falsify the axioms as stated, or force a measurable anisotropy of the ensemble.

6 Data availability

All results are reproducible from the scripts archived with this letter. `verificacion_letter.py` re-derives every numbered equation symbolically (sympy) and runs a 10^6 Monte Carlo smoke test; the canonical derivations are `derivacion_mu_angulo_solido_gates_direccionales.py` (the four gates of §3, exact), `derivacion_beta_ensamble_gates_reduccion_sympy.py` (the reduction of §4, exact via sympy), and `derivacion_beta_ensamble_gates_montecarlo_estable.py` (the 10^7 Monte Carlo of §4, verified to 50 digits with mpmath, with pre-declared acceptance bands and the anisotropic negative control; fixed seed). No proprietary data are used; the observational constraints cited are from the published SPARC literature. A record of the prior-literature search underlying the novelty claims of §5 is archived as `prior_work_verification.md`.

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